

MASSACHUSETTS GENERAL HOSPITAL

Performance Dashboard Documentation

1. Overview

- **Purpose:** The Healthcare Performance Dashboard is designed to provide comprehensive visibility into the operational and financial metrics of the healthcare facility. It tracks encounters, revenue over time, patient distributions.
- **Business Problem:** Hospitals often struggle with old, slow reports. This new automatic system solves this problem by giving clear, daily information. It helps top managers track the hospital's money, allows middle managers to learn about patient patterns, and helps daily staff organize doctors and beds at the right time.

2. Dataset

- **Data Sources:** The data is extracted directly from the hospital's internal systems and daily records.
- **Key Tables and Fields:**
 - encounters.csv: Contains 27,891 records of patient visits. Key fields include Id, START/STOP dates, PATIENT, PAYER, ENCOUNTERCLASS, TOTAL_CLAIM_COST, and PAYER_COVERAGE.
 - procedures.csv: Contains 47,701 records of medical procedures performed. Key fields include ENCOUNTER, CODE, DESCRIPTION, and BASE_COST.
 - patients.csv: Contains 974 patient demographic records. Key fields include Id, BIRTHDATE, RACE, GENDER, CITY, and STATE.
 - payers.csv: Contains 10 records of insurance providers (e.g., Medicare, Anthem).
 - organizations.csv: Contains facility information.
- **Data Size:** The dataset spans several years (up to 2022) with tens of thousands of transactional records, representing a robust sample for historical trend analysis.

3. Data Preparation

- **Cleaning Steps:**
 - Verified and handled any null values, particularly in fields like DEATHDATE (which is naturally null for living patients).
 - Standardized date and time formats in the START and STOP columns to enable accurate time-intelligence calculations.
 - Verified and correctly defined data types for all columns (e.g., setting cost columns to Currency and ensuring date fields are formatted as Date) to guarantee accurate calculations and filtering.
- **Transformations (Power Query):**
 - Calculated patient Age derived from BIRTHDATE and categorized them into functional Age Groups (e.g., Youth, Adults, Old, Dead).
 - Merged FIRST and LAST names to create a unified Full Name column for the details table.
 - Created a dynamic Calendar table to support advanced time-intelligence DAX functions (Year, Month, Day of Week).

4. Data Modeling

- **Schema Type:** The data model utilizes robust **Star Schema** architecture to ensure optimal performance and intuitive filtering in Power BI.
- **Fact Tables:** encounters sits at the center of the model as the primary fact table. procedures acts as a secondary detail/fact table linked directly to it.
- **Dimension Tables:** patients, payers, organizations, and the Calendar table surround the central fact table to provide descriptive context.
- **Relationships:**
 - Patients [Id] (1) to encounters [PATIENT] (*) - Primary to Foreign Key.
 - Payers [Id] (1) to encounters [PAYER] (*) - Primary to Foreign Key.
 - Encounters [Id] (1) to procedures [ENCOUNTER] (*) - Primary to Foreign Key.
 - Calendar [Date] (1) to encounters [START] (*) - Facilitates date-based slicing.

5. Calculations

The dashboard leverages robust DAX measures to calculate key performance indicators. Notable measures include:

- **Operational Measures:**
 - # Patients: Distinct count of patient IDs (DISTINCTCOUNT (patients [Id])).
 - # Encounters: Total number of visit records (COUNTROWS (encounters)).
 - # Emergency Encounters: Count of encounters where the class is 'Emergency'.
- **Financial Measures:**
 - \$ Total Revenue: Sum of all claim costs.
 - \$ Payer Coverage: Total amount covered by insurance providers.
 - \$ Patient Out of Pocket: Calculated as (\$ Total Revenue - \$ Payer Coverage).
 - % Coverage Rate: (\$ Payer Coverage / \$ Total Revenue).
- **Time-Intelligence Measures:**
 - \$ Total Revenue YoY
 - % Total Revenue YoY: Evaluates year-over-year financial growth.

6. Dashboard Layout

The interactive Power BI dashboard is structured into five distinct pages for targeted analysis:

- **Global Elements:** Every page features a top navigation bar, global slicers (Year, Age Group, RACE, GENDER, ENCOUNTERCLASS), and high-level KPI cards (\$Total Revenue, #Patients, #Encounters, #Emergency Encounters).
- **Page 1: Executive Summary:** Provides a macro-level view using line charts to track the yearly financial and operational trends (Revenue and Encounters)
- **Page 2: Financial Position:** Dives into revenue drivers. Such as (ENCOUNTERCLASS, Top Procedures, and over time)
- **Page 3: Demographics:** Focuses on patient profiling. Such as (country, Race, Age Group, and Gender)
- **Page 4: Seasonality:** Analyzes encounters seasonality.
- **Page 5: Details:** A detailed table showing patient details, payer information, time spent, and total revenue. It uses simple visual bars to help you quickly find the highest amounts.

7. Conclusion

- **It Helps Managers by:**
 - **Top Managers:** They see where the hospital money comes from.
 - **Middle Managers:** They learn about the patients (age, city, gender).
 - **Daily Operations:** They see the busy times to plan for doctors, nurses, and beds.